



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to VASANTHA ENTERPRISES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

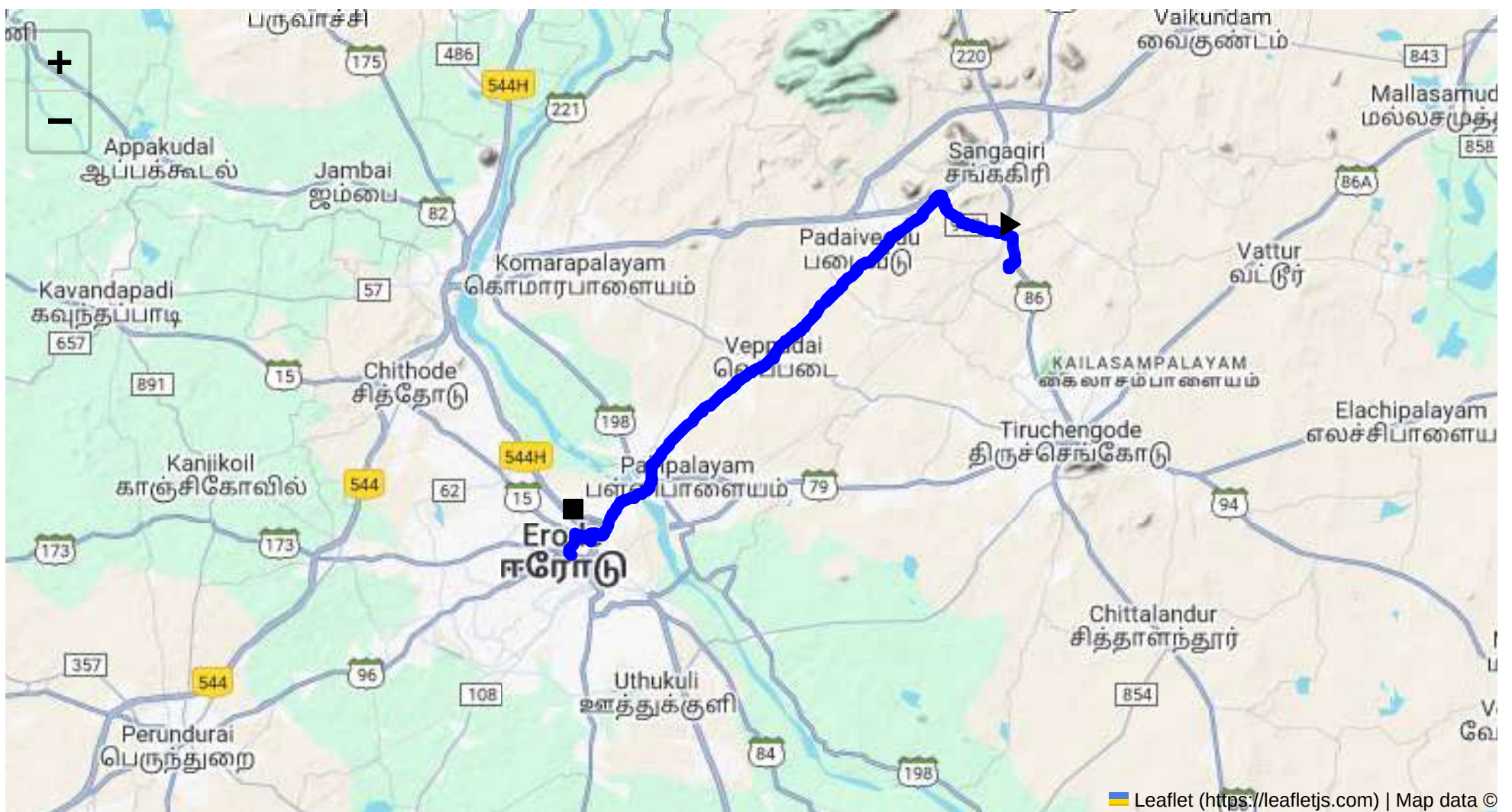
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 25.97 km**  
**Estimated Duration: 0.7 hours**  
**Adjusted Duration (Heavy Vehicle): 0.9 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.340379, 77.720093)**



## Welcome to the Journey Risk Management Study

### Route Overview

The route begins at Sangagiri, Tamil Nadu, and follows a path toward Erode, passing through key locations in Thirunagar Colony and Pavalatham Palayam before reaching Brough Road in Erode Fort. The total distance is approximately 26 kilometers and should take about 42 minutes under optimal conditions for heavy vehicles.

## Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. During the monsoon season (June to September and October to December), heavy rains could cause flooding, especially on poorly drained roads. During summer, extreme heat could lead to asphalt softening, posing challenges for heavy trucks.

## Traffic Patterns

Heavy congestion can be expected during peak hours (8:00-10:00 AM and 5:00-7:00 PM), particularly near urban areas such as Erode city centers and near significant intersections. This route may have moderate traffic during mid-day and evening hours, so planning travel post-rush hours can mitigate delays.

## Road Quality and Infrastructure

The roads on this route range from national highways to smaller roadways in urban settings. National highways feature good surface conditions and clear signage, while inner-city streets might have potholes and varying quality. Narrow streets and pedestrian traffic near urban areas can be a concern.

## Alternative Routes for Emergencies

In case of blockades or traffic issues:

- **Bypass Road:** Use NH544, which offers a more direct route avoiding central congestion at Brough Road.
- **Alternate Entry into Erode:** Access via the southern routes like Perundurai Road can circumvent city center congestion.

## Local Regulations on Hazardous Material Transport

Transporting hazardous materials requires strict adherence to permits in India, including permissible hours that typically avoid peak times and specific routes to minimize public exposure.

## Historical Incidents

There have been sporadic incidents involving vehicle mishaps in both rural and urban settings in Erode district, but no recent major accidents specifically involving hazardous materials have been reported.

## Environmental Considerations

The regions around Erode host several agricultural lands, which are sensitive to chemical runoff. Drivers must ensure no leakage of hazardous materials to avoid environmental harm.

## Communication Coverage

Cellular and GPS coverage are robust throughout Tamil Nadu, although some dead zones might be present in rural stretches, reducing certain telecom provider efficiency.

# Emergency Response Times

In urban and suburban regions like Erode, emergency response times are approximately 15-30 minutes owing to readily available local facilities. In more rural or remote areas, expect longer response times due to road access issues.

## Risk Assessment Summary

Overall, the route from Sangagiri to Erode is manageable but requires careful attention to weather conditions, traffic, and adherence to local regulations for hazardous materials. Effective journey planning and off-peak travel timing could ease potential risks. Ensure all contingency plans are activated for efficient communication and emergency response, especially when transporting hazardous materials.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.37086, 77.74828 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2  | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.45348, 77.85706 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.46312, 77.84886 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.34678, 77.73104 | 15 KM/Hr    |
| 11 | Blind Spot | Blind Spot | 11.34680, 77.72584 | 10 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.34492, 77.72706 | 10 KM/Hr    |
| 13 | Blind Spot | Blind Spot | 11.34716, 77.72174 | 10 KM/Hr    |
| 14 | Turn       | Medium     | 11.34477, 77.72149 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.34049, 77.71907 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------------|---------------------------|-------------|------------|
| 1  | hospital | Cheran Hospital                     | 11.3680302,<br>77.7481976 | 30 km/h     | Medium     |
| 2  | pharmacy | Pharmacy                            | 11.3652161,<br>77.7486375 | 30 km/h     | Medium     |
| 3  | hospital | Sri Kumaran E N T Hospital          | 11.3657814,<br>77.7477763 | 30 km/h     | Medium     |
| 4  | hospital | Sri Krishna Hospital                | 11.3656648,<br>77.7477783 | 30 km/h     | Medium     |
| 5  | hospital | Visaalam Hospitals                  | 11.3651528, 77.747808     | 30 km/h     | Medium     |
| 6  | police   | Police Station                      | 11.3630177,<br>77.7476505 | 30 km/h     | Medium     |
| 7  | hospital | Government Hospital                 | 11.3608508, 77.747453     | 30 km/h     | Medium     |
| 9  | hospital | Sri Hari Sakthivel Hospital         | 11.3526901,<br>77.7323055 | 30 km/h     | Medium     |
| 10 | hospital | Senthil Prakash Hospital            | 11.352639, 77.732267      | 30 km/h     | Medium     |
| 11 | hospital | V K Hospital                        | 11.3513792,<br>77.7313976 | 30 km/h     | Medium     |
| 12 | hospital | Dhanvantri Critical Care Center     | 11.3504266,<br>77.7311274 | 30 km/h     | Medium     |
| 13 | hospital | Dhanvantri Critical Care Centre     | 11.35099, 77.730508       | 30 km/h     | Medium     |
| 14 | hospital | Indrajith Yoga Nature Cure Hospital | 11.3499973, 77.732974     | 30 km/h     | Medium     |
| 16 | hospital | Kumudha Hospital                    | 11.3467172,<br>77.7306797 | 30 km/h     | Medium     |
| 17 | hospital | City Hospital                       | 11.3451151,<br>77.7300887 | 30 km/h     | Medium     |
| 18 | clinic   | K.B.N. Nursing Home                 | 11.344835, 77.7310771     | 30 km/h     | Medium     |
| 19 | hospital | Kaviraj Siddha Hospital             | 11.347153, 77.720863      | 30 km/h     | Medium     |
| 20 | hospital | SG MET Hospital                     | 11.34467, 77.720405       | 30 km/h     | Medium     |
| 21 | pharmacy | Chinthamani Medical Department      | 11.347629, 77.7190874     | 30 km/h     | Medium     |
| 22 | clinic   | K G Nursing Home                    | 11.3456845,<br>77.7188784 | 30 km/h     | Medium     |
| 23 | pharmacy | Pasumai Pharmacy                    | 11.3452083,<br>77.7183883 | 30 km/h     | Medium     |
| 24 | hospital | Universal Hospital                  | 11.3411046,<br>77.7185217 | 30 km/h     | Medium     |
| 25 | hospital | Devi Hospital                       | 11.34116, 77.718307       | 30 km/h     | Medium     |

|    | type     | name                                    | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------|---------------------------|-------------|------------|
| 26 | hospital | Abisek Medical Centre                   | 11.3415975,<br>77.7191789 | 30 km/h     | Medium     |
| 27 | hospital | SCS Hospital And Urology Clinic         | 11.3410253, 77.718043     | 30 km/h     | Medium     |
| 28 | clinic   | Vanalikaa Homoeo Centre                 | 11.341349, 77.717987      | 30 km/h     | Medium     |
| 29 | clinic   | Skin Care Clinic                        | 11.3409042,<br>77.7178413 | 30 km/h     | Medium     |
| 30 | hospital | Shri Vijaiah Trauma Care Hospital       | 11.3418935,<br>77.7193036 | 30 km/h     | Medium     |
| 31 | pharmacy | Tava Drug Mart                          | 11.3436578,<br>77.7182673 | 30 km/h     | Medium     |
| 33 | hospital | Dr. Agarwal's Eye Hospital Ltd. - Erode | 11.3421842,<br>77.7173066 | 30 km/h     | Medium     |
| 34 | clinic   | Janet Nursing Home                      | 11.3396455,<br>77.7203698 | 30 km/h     | Medium     |
| 35 | clinic   | Dr. R Kumaraswamy Sexologist Clinic     | 11.3396057,<br>77.7221701 | 30 km/h     | Medium     |
| 36 | hospital | Sri Vijaya Hitech Hospital              | 11.3400011,<br>77.7192251 | 30 km/h     | Medium     |
| 37 | hospital | Suriya Hospital                         | 11.340248, 77.7186452     | 30 km/h     | Medium     |
| 38 | hospital | Abitha Eye Hospital                     | 11.34021, 77.7187589      | 30 km/h     | Medium     |
| 39 | hospital | SRC Hospital                            | 11.3413999,<br>77.7177834 | 30 km/h     | Medium     |
| 40 | hospital | Guhan ENT Hospital                      | 11.339489, 77.721001      | 30 km/h     | Medium     |
| 41 | hospital | Surya Hospital                          | 11.3400026,<br>77.7190977 | 30 km/h     | Medium     |
| 42 | clinic   | Dr. Chandran Skin Laser Centre          | 11.3413527,<br>77.7174891 | 30 km/h     | Medium     |
| 43 | hospital | District Hospital, Eroda                | 11.3393048, 77.717926     | 30 km/h     | Medium     |
| 44 | hospital | Duraisamy Hospital                      | 11.3391625,<br>77.7187054 | 30 km/h     | Medium     |
| 45 | hospital | Anusha Hospital                         | 11.3397168,<br>77.7166805 | 30 km/h     | Medium     |
| 46 | hospital | Dr. Pondiperumal Hospital               | 11.3390924,<br>77.7205408 | 30 km/h     | Medium     |
| 47 | hospital | Ortho Life Hospital                     | 11.3395729,<br>77.7167329 | 30 km/h     | Medium     |
| 48 | hospital | Nalan Hospital                          | 11.3399703, 77.716773     | 30 km/h     | Medium     |
| 49 | hospital | Erode Kidney's Centre                   | 11.3383801,<br>77.7164201 | 30 km/h     | Medium     |

|    | type     | name                          | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------|---------------------------|-------------|------------|
| 50 | hospital | Srimathi Murugesan Hospital   | 11.3389205,<br>77.7187791 | 30 km/h     | Medium     |
| 51 | hospital | Care Hospital                 | 11.338443, 77.716511      | 30 km/h     | Medium     |
| 52 | clinic   | Sathyavathi Ramalingam Clinic | 11.3380366,<br>77.7166875 | 30 km/h     | Medium     |
| 53 | clinic   | Newlife Fertility Centre      | 11.3389396,<br>77.7213358 | 30 km/h     | Medium     |
| 54 | hospital | Sri Hari Hospital             | 11.338357, 77.7164204     | 30 km/h     | Medium     |

Crowded Spots

|    | type        | name                                       | coordinates               | speed_limit | risk_level |
|----|-------------|--------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | Ssri Valliappa Vidhayalayam mat hr school  | 11.4004878,<br>77.7794381 | 30 km/h     | Medium     |
| 8  | marketplace | Weekly Market                              | 11.3605669,<br>77.7488199 | 30 km/h     | Medium     |
| 15 | school      | Municipality Girls Higher secondary school | 11.3495618,<br>77.7325011 | 30 km/h     | Medium     |
| 32 | school      | Sengunthar matriculation School            | 11.3407075,<br>77.7195086 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High

**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.37086, 77.74828**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**  
**Risk Level: Medium**  
**Speed Limit: 30 KM/Hr**  
**Coordinates: 11.44898, 77.87410**



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45348, 77.85706



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46312, 77.84886



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.34678, 77.73104



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.34680, 77.72584



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.34492, 77.72706



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.34716, 77.72174



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.34477, 77.72149



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.34049, 77.71907

